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from tensorflow.keras.datasets import fashion_mnist
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# 1. Load Fashion MNIST dataset
(X_train, y_train), (X_test, y_test) = fashion_mnist.load_data()

# 2. Use a smaller dataset for faster KNN
X_train = X_train[:10000]
y_train = y_train[:10000]

X_test = X_test[:2000]
y_test = y_test[:2000]

# 3. Convert 28x28 images into 1D vectors
X_train = X_train.reshape(len(X_train), -1)
X_test = X_test.reshape(len(X_test), -1)

# 4. Normalize pixel values
X_train = X_train / 255.0
X_test = X_test / 255.0

# 5. Try different K values
for k in [1, 3, 5, 7]:
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    print("K =", k, "Accuracy =", accuracy)
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K = 1 Accuracy = 0.8075
K = 3 Accuracy = 0.8195
K = 5 Accuracy = 0.8225
K = 7 Accuracy = 0.826
```